Klara Winther

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SUMMARY

Data Analyst and Ecologist experienced in cloud-based geospatial pipelines. Skilled in spatial data analysis, including the use of Python, SQL, GIS, Github, and Microsoft Azure to produce large-scale landcover maps in a collaborative environment. Proud of having participated in full map-making pipelines from reviewing and synthesising scientific literature and providing ecological rationale to handling complex geospatial datasets and transforming them into insightful visual outputs.

TECHNICAL SKILLS

- **Programming languages:** Python, SQL (PostgreSQL), R, JavaScript
- **Python libraries:** Numpy, Dask, Pandas, GeoPandas, xarray, rioxarray, Rasterio
- **Geospatial tools:** ArcGIS, ArcGIS Online, QGIS, Google Earth Engine, Collect Earth Online
- **Cloud & data platforms:** Microsoft Azure
- **Web & version control:** HTML, CSS, Git, GitHub, Dagster

EMPLOYMENT

Space Intelligence, Edinburgh, Scotland
Ecologist / Data Analyst (Mapping Quality Expert) *Dec 2024 – Present*
Ecologist Intern *Jul 2024 – Dec 2024*

- Contributed to the delivery of 20+ large-scale landcover map projects, functioning both as ecologist and data analyst, thus partaking in the full map-making pipeline.
- Took ownership of point-based accuracy assessments of large-scale landcover maps, managing the setup independently across **Python**, **Collect Earth Online** and **Azure blob storage**.
- Managed interns and other team members, helping with technical setup in **QGIS**, explaining workflows and delegating tasks related to map quality assurance and point-based accuracy assessments.
- Independently initiated a database schema in **QGIS** for a Scotland-wide landcover map, using **Python** and **PostgreSQL** to create and manage database tables.
- Processed complex geospatial datasets in Python using **Geopandas**, **Rasterio** and **Xarray**, and produced several intermediary landcover and change map outputs as zarr and raster data.
- Worked on cloud-based geospatial data pipelines within **Microsoft Azure**, using **Dask** for parallel computation and **Dagster** for structured workflow orchestration.
- Transformed geospatial data into insightful plots in Python using **Matplotlib** and **Seaborn** as well as creating landcover class dictionaries to stylise maps in QGIS.
- Conducted version control using **Git**, including committing pull requests on company-wide Github repository templates.
- Resolved 50+ bugs in the process of setting up Python environments and running notebooks in **VS-Code**, contributing to a smoother workflow.

- Improved SOPs by reviewing ecological and technical workflows, synthesising scientific literature and project evidence, and refining definitions to support consistency and reproducibility.

FIELDWORK EXPERIENCE

Field Assistant, University of Edinburgh, Drumbrae/Ochil Hills *April 2024 - May 2024*

- Conducted structured soil sampling across multiple sites to assess carbon content and storage capacity under differing land-use scenarios.
- Excavated soil pits and collected stratified samples and soil cores in accordance with research protocols.
- Recorded samples for laboratory analysis, ensuring traceability and consistency of metadata.

Field Assistant, University of Cambridge, Norfolk Broads *August 2023*

- Carried out structured vegetation surveys within $20 \times 20\ m$ plots in forested peatland environments.
- Identified, measured and tagged trees, recording **DBH** and ecological attributes.
- Collected high-accuracy positional data ($<30\ cm$) to align ground observations with **TLS** and drone **LiDAR** datasets.

EDUCATION

University of Edinburgh, Edinburgh, Scotland
MA Geography, *First Class Honours* *September 2019 - May 2024*

- Evaluated the performance of spectral indices for classifying peatland condition across Scotland using Sentinel-2 imagery.
- Implemented prepocessing workflows across **Google Earth Engine** and **ArcGIS** using **Python** and **JavaScript** to construct a Scotland-wide spatio-temporal dataset of surveyed peatland sites.
- Trained and executed a supervised classification **(Random Forest)** model, including conducting spectral index derivation, hyperparameter tuning and running classification accuracy metrics.

PROFESSIONAL TRAINING & CERTIFICATION

HTML Tutorial; CSS Tutorial; JavaScript Tutorial *W3Schools* *February 2026 - Present*

Using Spatial Data for Biodiversity, *UNDP & NASA* *May 2023*

Introduction to Ecosystem Restoration, *UNDP* *May 2023*

Biodiversity Applications for Airborne Imaging Systems, *NASA* *March 2023 - April 2023*

Going Places with Spatial Analysis, *ESRI* *February 2023 - March 2023*

Technical skills/applications used: Pandas, GeoPandas, GDAL, Numpy, R, ArcGIS Online, UN Biodiversity Lab, HTML, CSS, JavaScript.

References upon request.